Lists, in practice

TC1037 · Algorithmic Thinking · Session 12 · max, min and mean over a column

---

Dr. Leopoldo Rodríguez Salazar

Friday 6 November 2026

Tecnológico de Monterrey · Campus Querétaro · Group 510

First, an apology — and a standing invitation

I cancelled tutoring sessions without telling you.

Urgent work came up, I dealt with it, and I lost track of the clock. That is on me, not on you, and it is not how I want this to run.

If you were one of the people left waiting: I am sorry.

Next week I am open.

Book a slot — the link is in the Canvas announcement.

Bring something specific: your midterm paper, the code that will not run, your dataset. Fifteen minutes on a real problem beats an hour of nodding.

Due dates — tonight, Sunday, Tuesday

When	What	What counts as done
Tonight 23:59	**Deliverable 1** `entrega1.xlsx`	An Excel table from a real source. **30–50 rows**, up to **10 columns: 5 text 5 numeric**. In pairs. One upload pe team.
Sunday 23:59	**Homework 6 + Lab 6**	Lists. Both on Canvas now.
Tuesday, in class	**Quick exam 5**	Lists. Same format as the other four.

As of this morning: zero of you have uploaded Deliverable 1. Each missed or late stage costs **5 points** off the project — and the project is 15 % of the course.

Deliverable 1 — the whole thing, on one slide

Where to get data

kaggle.com/datasets · datos.gob.mx · datacatalog.worldbank.org

Or a real company that lets you use its numbers — **a family business counts**, and is often the better project.

The shape

30–50 rows. Up to 10 columns: about five text, about five numeric. You need both kinds — the text columns are what you will group by, the numeric ones are what you will average and plot.

How to not waste the weekend

1. Pick a subject you actually want an answer about.
2. Download the table. Open it. *Look* at it.
3. Delete the columns you will never ask about. Ten is a ceiling, not a target.
4. Save it as `entrega1.xlsx` and upload it as a pair.

Next Friday you write the questions you will ask this table. Pick data that has questions in it.

Where we stopped on Tuesday

```
speeds = [100, 90, 140, 80]

speeds[0]          # 100
speeds[3]          # 80
len(speeds)        # 4
speeds[len(speeds) - 1]    # 80
speeds.append(60)
```

One name, many values, **in order**.

Counting starts at **0**, so the last index is `len(xs) - 1` — and `speeds[4]` on a four-item list is an `IndexError`, not an empty box.

`append()` is how a list grows *inside* a loop.

Today: the three things you will do to every column of your dataset.

Three questions, one shape

```
def mean(xs):
    total = 0
    for x in xs:
        total = total + x
    return total / len(xs)

def maximum(xs):
    best = xs[0]
    for x in xs:
        if x > best:
            best = x
    return best
```

Same four beats every time:

1. start an accumulator
2. walk the list
3. update it
4. `return` it — outside the loop

Minimum is `maximum` with `<`. Write it in thirty seconds and you have three of the five statistics your project needs.

Two traps, both of them yours already

The return inside the loop

```
for x in xs:
    total = total + x
    return total    # <- stops
```

It returns after the *first* value. One indentation level decides whether you get the answer or the first item.

best = 0

```
best = 0
for x in temperatures:
    if x > best: best = x
```

Fine until a column is all negative — then the maximum of `[-3, -8, -1]` comes back `0`, a number that is not in your data. Start from `xs[0]`, not from zero.

Both of these cost marks on the midterm. The second one is the `n == 0` case wearing a different hat: *what does this do on the awkward input?*

Your turn — fifteen minutes, in pairs

```
ventas = [120, 340, 90, 560, 210,
          480, 75, 300]

# 1. total and mean
# 2. the largest and the smallest
# 3. how many are above the mean
# 4. a NEW list with only those
```

Rules, as always

- one function each, each one `return`s
- no `print()` inside a function
- `main()` calls them and labels every number it shows
- run the file *before* you call me over

Number 4 is `append()` inside an `if` inside a `for`. It is also exactly how you will filter a column of your dataset.

Why this is the project, not an exercise

```
# entrega1.xlsx, one column
ventas = [120, 340, 90, 560]

def estadistica_descriptiva(xs):
    return (mean(xs), maximum(xs)
    ,
            minimum(xs), len(xs),
            total(xs))
```

The final program needs **five descriptive statistics** over your data. You have just written them.

In week 9 pandas hands you the column instead of you typing it. Nothing else changes — same functions, same `main()`, same labels.

That is why tonight's upload matters: without a table, there is no column, and without a column there is nothing to run this on.

A list is a column.

A loop is a question you ask it.

Tonight 23:59 — `entrega1.xlsx`, in pairs

Sunday 23:59 — Homework 6 and Lab 6

Tuesday — Quick exam 5, in class

And book a tutoring slot. I am free next week and the link is in Canvas.